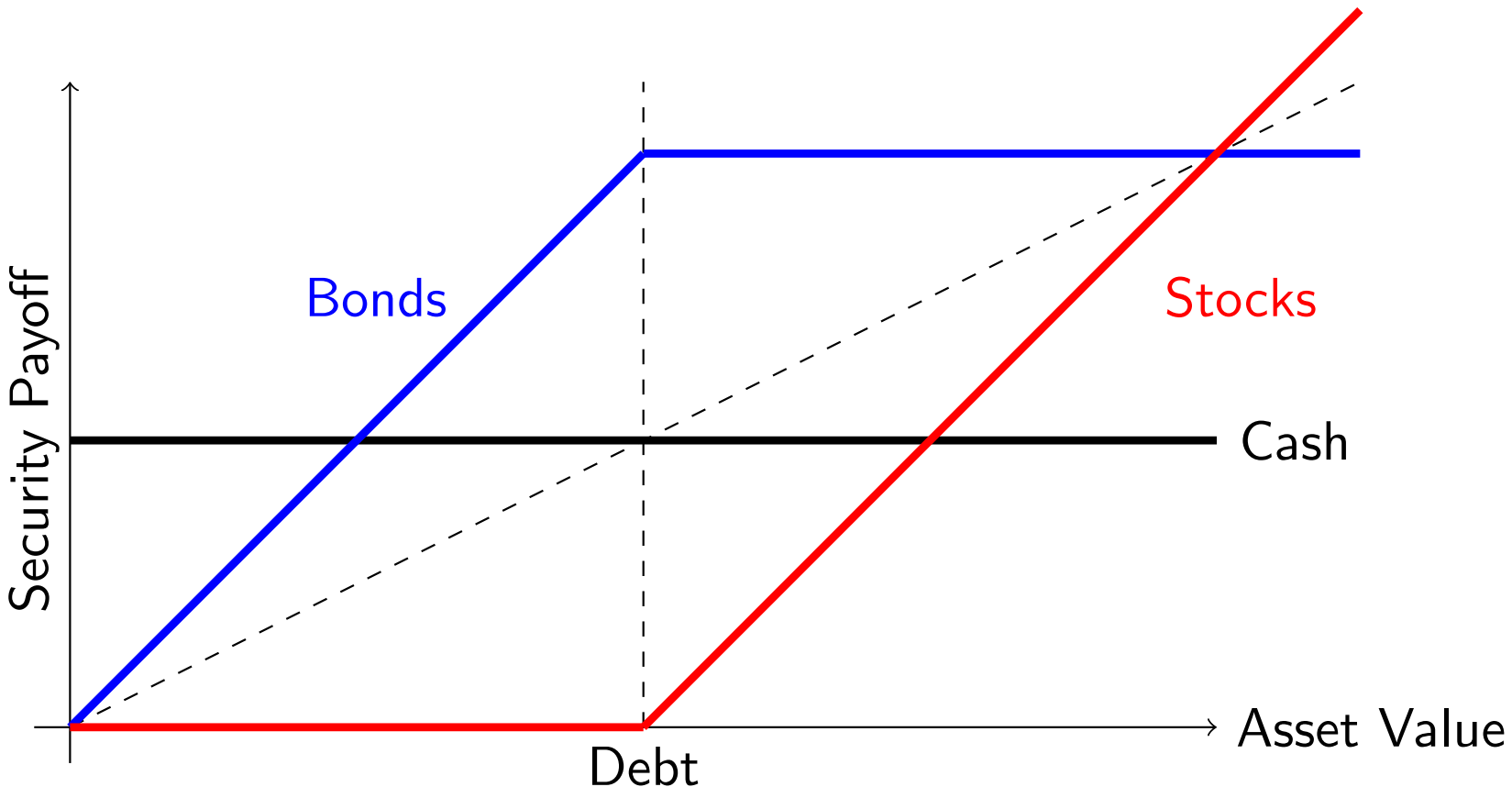
Asset Classes

January 27, 2020

Cash vs. Bonds vs. Stocks: An Asset-Based Perspective

- Remember that assets produce something of value, while securities are claims to the output of the asset
- The value of a firm's assets is uncertain, and depends on how well the firm performs, the state of the economy, and so on
- Suppose an investor's portfolio contains some cash, some of the firm's bonds, and some of its stock
- How does the value of her portfolio vary with the value of the firm's assets?

Cash vs. Bonds vs. Stocks: An Asset-Based Perspective



Example

- The investor has a cash balance of 10m USD and owns 40% of the firm's stock and 60% of its bonds
- The firm owes 50m USD (principal and interest) to bondholders and its asset are worth 75m USD
- How much is her portfolio worth?

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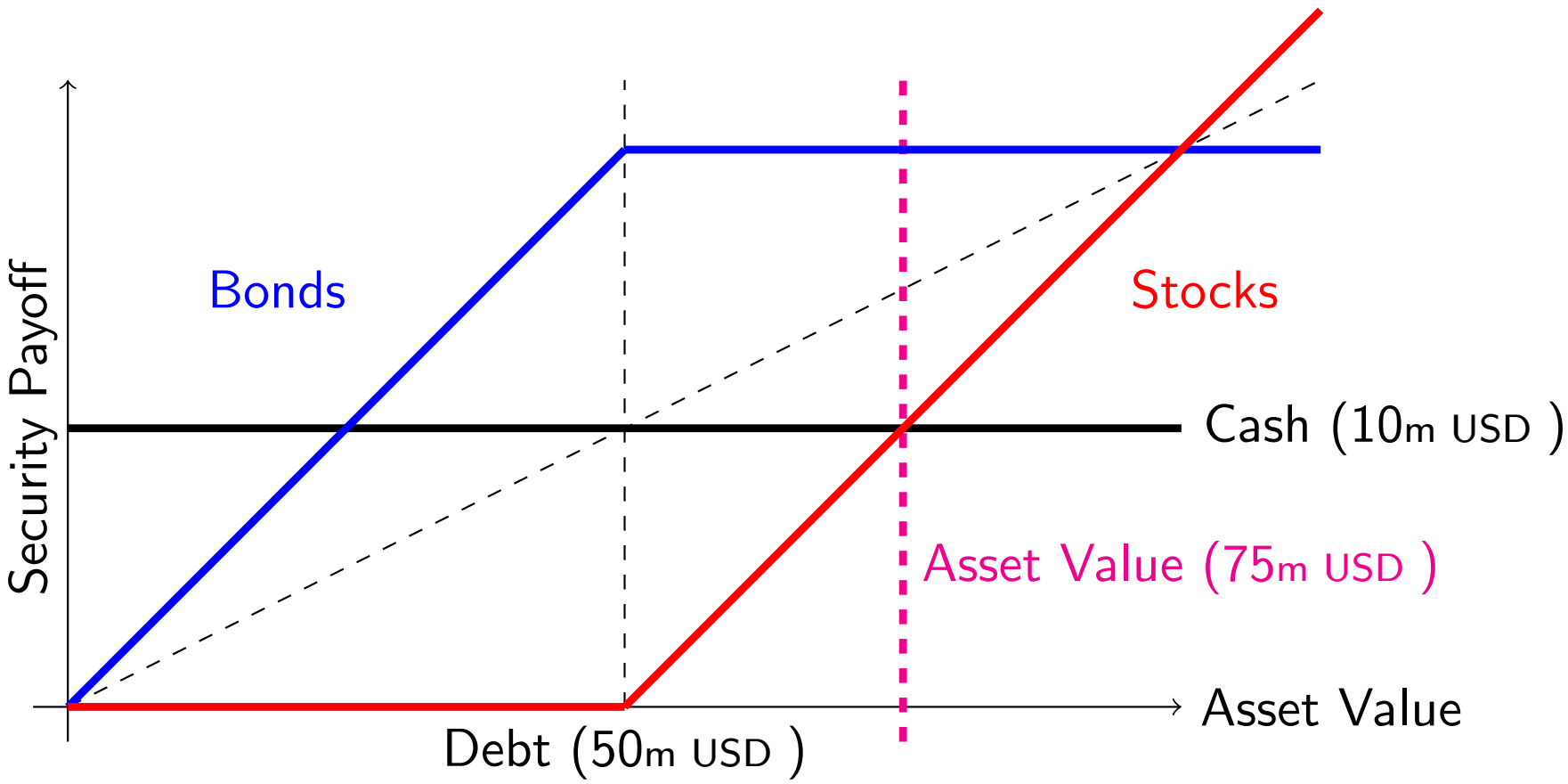

Example

- The firm pays 50m USD to bondholders, and the remaining 25m USD to stockholders
- The investor keeps her cash balance of 10m USD and receives 60%×50m USD =30m USD from the firm's bonds and 40%×25m USD =10m USD from the firm's stock
- Her portfolio is worth 50m USD

Example

- The investor has a cash balance of 10m USD and owns 40% of the firm's stock and 60% of its bonds
- The firm owes 50m USD (principal and interest) to bondholders and its asset are worth 25m USD
- How much is her portfolio worth?

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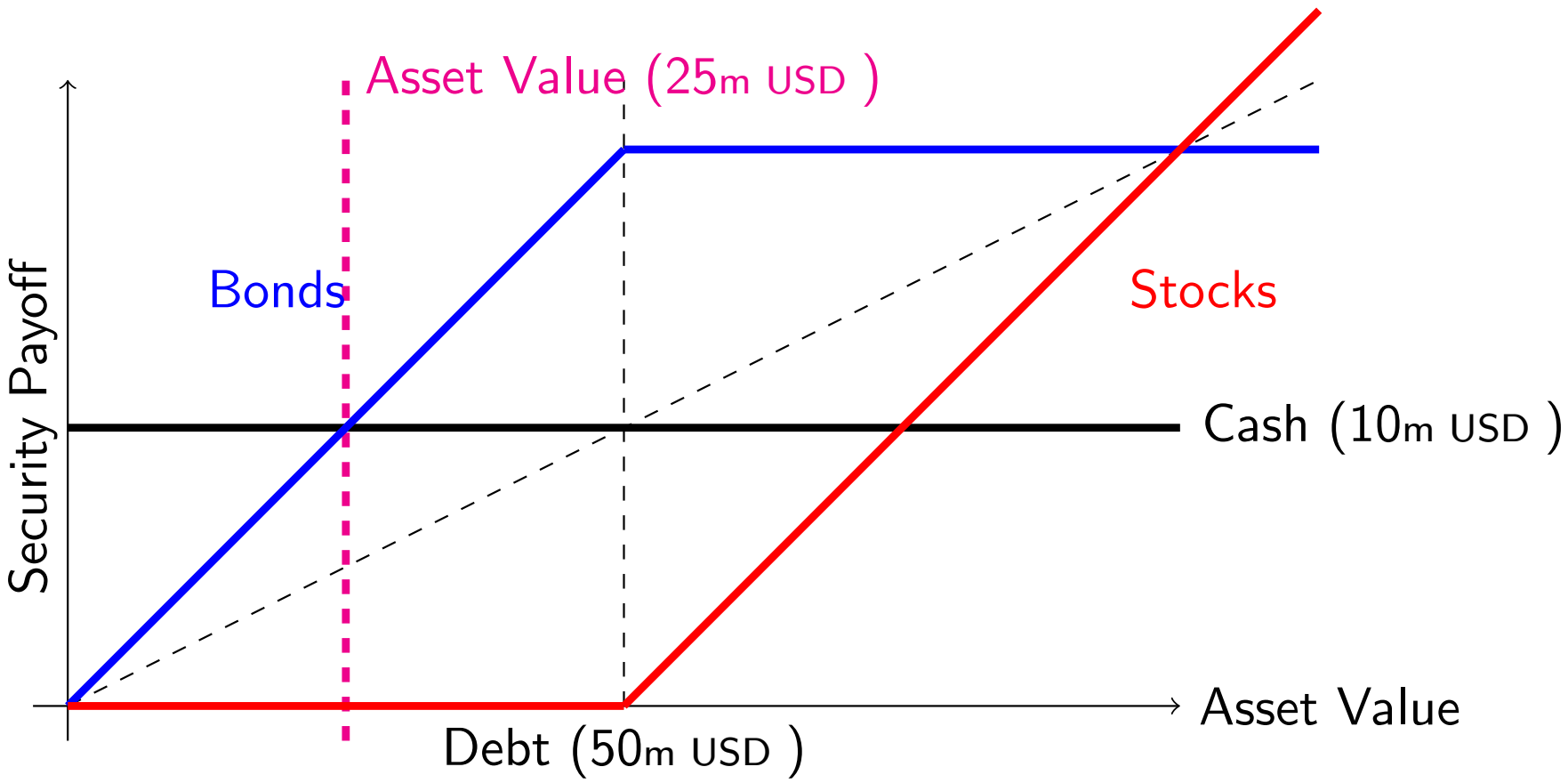


Example

- The firm pays 25m USD to bondholders and the stockholders get nothing
- The investor keeps her cash balance of 10m USD and receives 60%$\times$25m USD =15m USD from the firm's bonds and 40%$\times$0m USD =0m USD from the firm's stock
- Her portfolio is worth 25m USD

Cash (a.k.a. The Money Market)

- The money market is a subset of the bond market.
- These are short-term, highly liquid securities.
- They are often called *cash* or *cash equivalents*.

Treasury Bills[1]

- Most marketable of all money market instruments
- Investors buy the bills at a discount from the face value
- At maturity, the government pays the investor the face value of the bill
- The difference between the purchase price and the face value is the investor's earnings
- T-bills are issued with maturities of 4, 13, 26, and 52 weeks
- T-bills are highly liquid
- The income earned on T-bills is taxable at the federal level, it is exempt from state and local taxes

[1] Language from BKM

Miscellaneous

- Certificates of Deposit
  - A Certificate of Deposit (CDs) is a time deposit with a bank
  - They may not be withdrawn on demand
  - They are insured by the FDIC and may be traded on a secondary market
- Commercial Paper
  - Large corporations issue short-term notes to finance working capital
  - They have maturities less than 270 days to avoid registration with the SEC
  - They were historically used by non-financial firms, but notoriously used by financial firms just before the crisis
- Repos and Reverses
  - Short-term sales of securities with promise to repurchase at higher price
  - RP is a collateralized loan

Interest Rates

- Federal Funds
  - Banks are required to hold reserves at the Federal Reserve
  - Banks can trade their reserves with each other
  - Key interest rate for economy
- London Interbank Offered Rate (LIBOR)
  - Second only to Treasuries rates in importance
  - It is the rate at which banks lend to each othe
  - It is used as a benchmark rate for trillions of dollars worth of securities (primarily derivatives)

The Libor Scandal

My thwarted attempt to tell of Libor shenanigans

Douglas Keenan

In 1991, I began trading for Morgan Stanley, the investment bank, in London. I was trading bonds, derivatives and related securities. One of those securities was based on the three-month Libor rate: the interest rate at which banks can borrow money for three months from each other. Morgan Stanley does not trade on the interbank market so I could not directly borrow or loan money at Libor rates. What I could do, however, was trade a futures contract on the three-month Libor rate.

As an example of how a futures contract works, consider the following. Suppose that we are concerned about three-month Libor rates increasing in the future; in particular, we are concerned about what the three-month rate will be in September. If that rate is, say, 1 per cent, we can agree today to effectively lock it in. If, come September, the actual three-month rate is 2 per cent, then our contract will ensure we can still borrow at 1 per cent. Futures contracts on three-month Libor were – and are – traded on the London International Financial Futures Exchange (Liffe, now part of NYSE Euronext). There was a standard contract for the month of September. That contract had its rate settled on the third Wednesday of the month, at 11am.

In 1991, I had live trading screens that showed the Libor rates. In September of that year, on the third Wednesday, at 11am, I watched those screens to see where the futures contract *should* settle. Shortly afterwards, Liffe announced the contract settlement rate. Its rate was different from what had been shown on my screens, by a few hundredths of a per cent.

As a result, I lost money. The amount was insignificant for me, but I believed that I had been defrauded and I complained to Liffe. Liffe explained that the settlement rate was not determined by what rates were actually in the market. Instead, the British Banker's Association polled some banks, asking them what the rates were. The highest and lowest reported rates were discarded and the rest averaged, giving the settlement rate. Liffe explained that, in doing this, they were adhering to the terms of the contract.

I talked to some of my more experienced colleagues about this. They told me banks misreported the Libor rates in a way that would generally bring them profits. I had been unaware of that, as I was relatively new to financial trading. My naivety seemed to be humorous to my colleagues.

Simply put, then, it seems the misreporting of Libor rates may have been common practice since at least 1991. Although the difference between the reported rate and the actual rate might seem small, the total amount of money involved is material, given that Libor rates affect contracts worth hundreds of trillions. Also important is what such misreporting says about the culture.

I had been unaware of Libor misreporting, as I was new to trading. My naivety seemed humorous to colleagues

During 1991, at the London office of Morgan Stanley, the head of interest rate trading was a person who has been at the centre of the current scandal: Bob Diamond. I do not recall discussing Libor misreporting with Mr Diamond but since the misreporting was common knowledge among traders, I presume he was aware. (That, however, is not a criticism of Mr Diamond: what could he have done about this?)

There have been two distinct motivations for banks to misreport Libor rates. One motivation is discussed above: to directly increase profits. The other motivation arose during the 2008 financial crisis: to mask liquidity problems.

Libor misreporting has been going on for decades. Why have investigations only recently begun? It seems highly implausible that all the investigating agencies could have been unaware for decades. Indeed, those agencies have a reputation among traders of being like Potemkin villages. I suspect what has happened is that, after the financial crises of 2008, the agencies decided they ought to perform more of their stated duties. That would also explain why the investigations appear to be ignoring any misreporting in years before 2005: to cover up the illusoriness of their earlier work.

One of the investigations is being undertaken by the House of Commons Treasury committee. I telephoned the committee on July 3 and spoke with a committee specialist. I told the specialist about the foregoing and said that I was willing to testify under oath. The specialist seemed extremely interested. They said they were to have a meeting about the Libor scandal and would call me back afterwards. I did not hear back, however, so I phoned to ask what was happening. My testimony was not wanted, the specialist told me, because it "contradicts the narrative".

The writer is an independent mathematical scientist and a former Morgan Stanley trader

- Before 2012, LIBOR was literally a poll.
- Banks submitted what they "thought" they could borrow/lend at with other banks.
- It was discovered that banks were manipulating the poll to favor their trading positions.

Government Debt

- Treasury Notes and Bills
  - Notes: up to 10 year maturities
  - Bonds: up to 30 year maturities
- Treasury Inflation Protected Securities (TIPS).
  - Principal adjusted for changes in the CPI
  - Marked with a trailing "i" in quote sheets
- Municipal Bonds.
  - Issued by state and local governments to finance capital improvement projects (e.g. roads, schools, parks).
  - Interest income is exempt from state and local taxation.
- Federal Agency Debt.
  - Bonds issued by Government Sponsored Enterprises (GSEs) to finance purchase of consumer debt
  - Most are home-mortgage-related: FNMA ("Fannie"), FHLMC ("Freddie"), GNMA ("Ginnie")

Miscellaneous

- Corporate Bonds.
  - You know a lot about these already.
  - Two new species:
    - Callable Bonds: The firm can buy back the bond at a specified price
    - Convertible Bonds: The bondholder can convert her bond into a specified number of shares of common stock
- International Bonds.
  - Bonds issued in another currency
  - E.g. Yankee bond is a bond sold in the US in USD by a non-US issuer; Samuari bond is a bond sold in Japan in Yen by a non-Japanese issuer, etc.

Asset Backed Securities (ABS)

- Put a bunch of securities in a portfolio called a "pool"
- Create securities out of equity shares in the pool
- Voila! You've made an asset backed security
- Examples of assets turned into ABSs:
  - Mortgages ("Mortgage Backed Securities")
  - Life insurance policies ("death bonds")
  - Credit card debt
  - Student debt
  - Receivables
  - Home equity loans
  - Auto loans

Indices

- United States
  - Dow Jones
  - S&P 500
  - NASDAQ composite
- Europe (and East Asia)
  - FTSE (UK)
  - CAC (France)
  - DAX (Germany)
  - Nikkei (Japan)
- China
  - Shanghai Composite
  - Hang Seng (Hong Kong)

Securities

- Common Stock
  - You know a lot about these already
- Preferred Stock
  - Get paid before common stockholders, but can't vote
  - The firm has some discretion in paying preferred stock holders
- American Depositary Receipts
  - Certificates traded in the U.S. representing ownership in foreign security (e.g. Sony trades on the NYSE under the ticker SNE)

Exchange Traded Funds

- An Exchange-Traded Fund (ETF) is a fund that allows investors to trade an index portfolio (e.g. S&P 500)
- The ETF trades just like a stock and has its own ticker (e.g. SPDR tracks the S&P 500)
- Benefits (relative to traditional mutual funds)
  - Trade continuously throughtout the day
  - Can be sold or purchased on margin
  - Potentially lower tax rates
  - Lower costs (no marketing, lower fund expenses)
- Net Asset Value (NAV) is a fund's assets (the securities in its portfolio) minus liabilities (debt financing or other obligations)
- ETF price can deviate from the NAV (tracking error)

Indices

- An <u>index</u> is the value of a chosen portfolio
- Dow Jones Industrial Average (DIJA)
  - 30 large stocks
  - Price weighted average
- Standard & Poor's (S&P 500)
  - 500 largest stocks
  - Value weighted average
- Specialized indices (e.g. FAANG)

Why do we have indices?

Return on Value-Weighted Index (E.g. S&P 500)

- N stocks, $j = 1, 2, \ldots, N$; $p_{j,t}$ price of stock j on date t; $r_{j,t}$ total return on stock j from date t to date $t+1$; n_j number of shares of stock j
- $n_j p_{j,t}$ is j's market capitalization (cap) on date t
- The level of the index is

$$V_t = n_1 p_{1,t} + n_2 p_{2,t} + \cdots + n_N p_{N,t} \tag{1}$$

- The *return* of the index is

$$\frac{n_1 p_{1,t}}{V_t} r_{1,t} + \frac{n_2 p_{2,t}}{V_t} r_{2,t} + \cdots + \frac{n_N p_{N,t}}{V_t} r_{N,t} \tag{2}$$

- **Important:** the rate of return on a value-weighted index is the rate of a return on a portfolio that buys-and-holds all of the shares of all of the securities that make up the index

Example

The following data are from 2015-08-01. What is the value-weighted return of the portfolio?

Stock	**Cap**	**Return**
American (AAL)	26.1876	-0.91%
Alaska (ALK)	9.5313	1.59%
Delta (DAL)	34.8225	-0.38%
Southwest (LUV)	24.1984	1.03%
United (UAL)	21.5211	-0.96%

market cap in millions of USD, returns monthly total returns

Example

To find the value-weighted return...

Stock	**Cap**	**W**eight	**R**eturn	**W**×**R**
American (AAL)	26.1876	0.2252	-0.91%	-0.205%
Alaska (ALK)	9.5313	0.082	1.59%	0.1304%
Delta (DAL)	34.8225	0.2995	-0.38%	-0.1138%
Southwest (LUV)	24.1984	0.2081	1.03%	0.2144%
United (UAL)	21.5211	0.1851	-0.96%	-0.1777%
Portfolio	116.2609	0.9999		-0.1517%

market cap in millions of USD, returns monthly total returns

Example

The following data are from 2015-03-01. What is the value-weighted return of the portfolio?

Stock	**Cap**	**Return**
American (AAL)	36.6188	3.97%
Alaska (ALK)	8.6609	2.59%
Delta (DAL)	37.0593	-8.33%
Southwest (LUV)	29.9465	3.18%
United (UAL)	25.8392	0.99%

market cap in millions of USD, returns monthly total returns

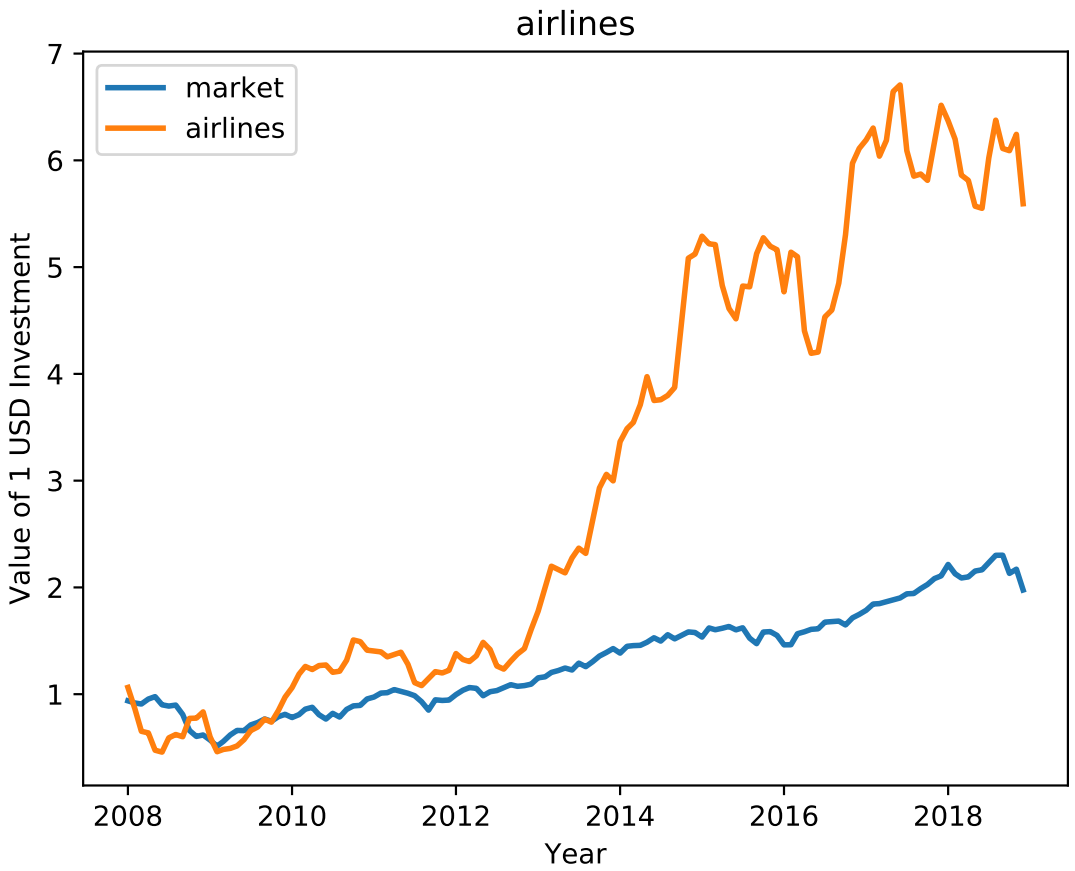
Example

To find the value-weighted return...

Stock	**Cap**	**W**eight	**R**eturn	**W**×**R**
American (AAL)	36.6188	0.2651	3.97%	1.0525%
Alaska (ALK)	8.6609	0.0627	2.59%	0.1624%
Delta (DAL)	37.0593	0.2683	-8.33%	-2.235%
Southwest (LUV)	29.9465	0.2168	3.18%	0.6894%
United (UAL)	25.8392	0.1871	0.99%	0.1852%
Portfolio	138.1247	1.0		-0.1455%

market cap in millions of USD, returns monthly total returns

From the Data



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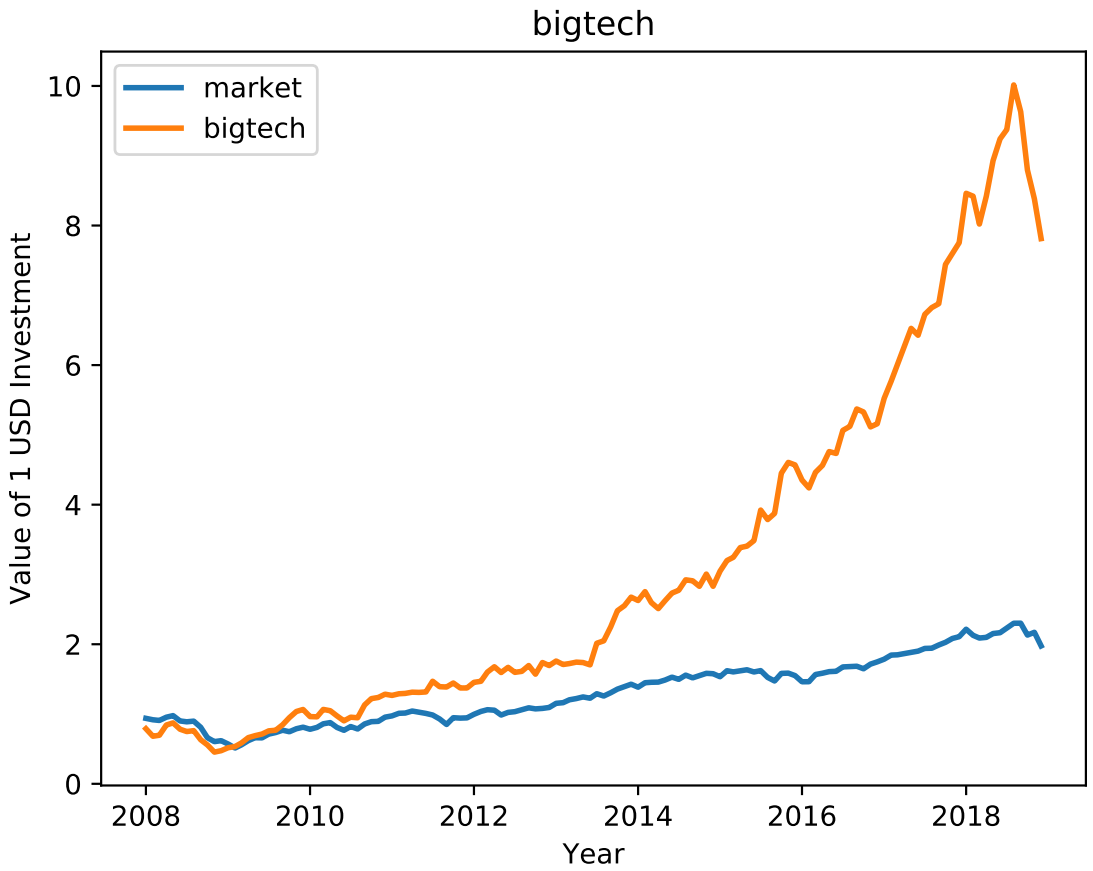

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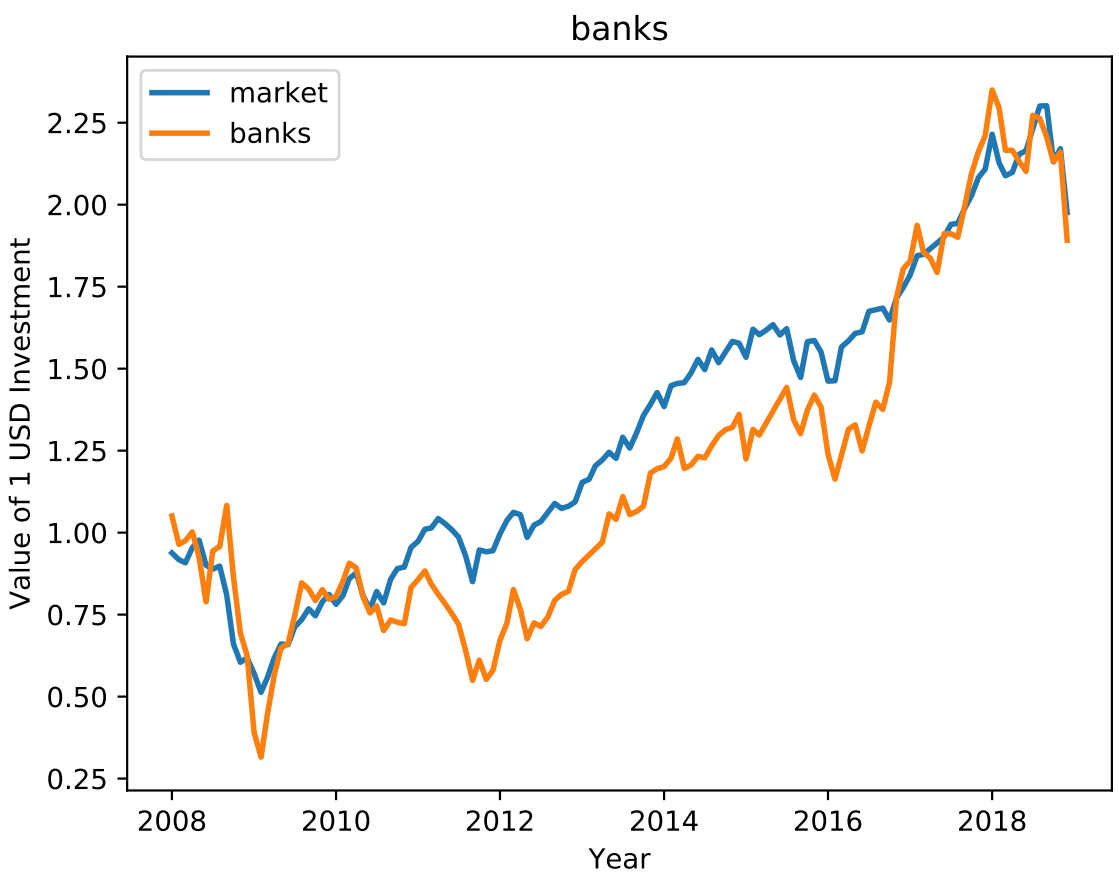

Return on Equal-Weighted Index

- N stocks; $r_{j,t}$ total return on stock j from date t to date $t+1$
- The *return* of the index is

$$\frac{r_{1,t} + r_{2,t} + \cdots + r_{N,t}}{N} \tag{3}$$

- Equal weighting useful in settings were market-caps don't exist
- Example: Markit CDX Indices for Credit Default Swap (CDS)

Example

The following data are from 2016-04-01. What is the equal-weighted return of the portfolio?

Stock	**Return**
American (AAL)	-14.14%
Alaska (ALK)	-0.43%
Delta (DAL)	-7.74%
Southwest (LUV)	-23.48%
United (UAL)	-14.41%

Example

To find the equal-weighted return...

Stock	**W**eight	**R**eturn	**W**×**R**
American (AAL)	1/5	-14.14%	-2.828%
Alaska (ALK)	1/5	-0.43%	-0.086%
Delta (DAL)	1/5	-7.74%	-1.548%
Southwest (LUV)	1/5	-23.48%	-4.696%
United (UAL)	1/5	-14.41%	-2.882%
Portfolio	1		-12.04%

Example

The following data are from 2018-07-01. What is the equal-weighted return of the portfolio?

Stock	**Return**
American (AAL)	3.88%
Alaska (ALK)	14.15%
Delta (DAL)	2.47%
Southwest (LUV)	15.14%
United (UAL)	10.4%

Example

To find the equal-weighted return...

Stock	**W**eight	**R**eturn	**W**×**R**
American (AAL)	1/5	3.88%	0.776%
Alaska (ALK)	1/5	14.15%	2.83%
Delta (DAL)	1/5	2.47%	0.494%
Southwest (LUV)	1/5	15.14%	3.028%
United (UAL)	1/5	10.4%	2.08%
Portfolio	1		9.208%